

Diana He

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EDUCATION

University of Waterloo

Sept. 2025 – Present

Bachelor of Applied Science in Mechatronics Engineering

3.7 GPA

TECHNICAL SKILLS

Design: SolidWorks, AutoCAD, Surface Modeling, FEA, CFD, GD&T, DFMA, Tolerance Stack-Up Analysis

Fabrication: Spot Welding, CNC, Sheet Metal Fabrication, Composite Manufacturing, Laser Cutting, Soldering, Mechanical-Electrical Integration, Rapid Prototyping

Programming: Python, C++, MATLAB

EXPERIENCE

Battery Lead and Aerodynamics Member

Sept. 2025 – Present

Midnight Sun Solar Car Team

- Designed module endplates and retainers to correct 6 mm of tolerance stack-up, resolving module fit issues; designed electrical mounts following DFMA principles, positioning boards for easy debugging access.
- Engineered main and inter-module busbars in SolidWorks, selecting SigmaClad for conductivity and aluminum 6061 for bending stiffness over a long unsupported span.
- Diagnosed 20+ battery modules for inconsistent welds caused by busbar oxidation; resolved them with a standardized sanding and IPA prep procedure bringing success rate up from 50% → 90%.
- Identified and redesigned a redundant wheel shield mold to consolidate it into one, reducing layup complexity and material use across 8 shields; built roof and tub rib sandwich panels via resin infusion.
- Fabricated and reinforced a fiberglass windshield mold through wet layups to withstand ~200 lb of thermoforming pressure; produced a windshield which passed competition scrutineering.

FRC Robotics Electrical Lead

Sept. 2023 – April 2025

Team: 7690 and 8349

- Engineered and implemented Team 7690's electrical wiring design within tight space constraints, prioritizing visibility and accessibility of lights and components to speed up diagnostics during competition.
- Diagnosed and repaired critical electrical and mechanical failures in high-pressure competitions, including a faulty voltage regulator, detached JST connectors, and stripped bolt holes.
- Built plexiglass clamp components using CNC and Carbide Create across 4 design revisions over 2 weeks, deburring finished parts for prototyping and testing.
- Achieved 10th/30 at Newmarket Qualifiers and 15th/50 in Science Division at Provincials.

PROJECTS

Solar Car Battery Thermal Model | *Python*

- Developed a lumped-parameter thermal model based on Midnight Sun's FSGP/ASC 2026 36S9P pack, solving 324 coupled ODEs via SciPy's implicit BDF integrator to predict peak cell temperature across a 9 hour race day.
- Derived pack current discharge from first-principles vehicle dynamics and GPS elevation data, producing a predicted SOC drawdown of 75% to 29% that closely mirrored team telemetry.
- Validated with energy-balance closure and Biot-number checks; lumped assumption holds at nominal load but becomes marginal at worst case.
- Created time-series visualizations of race-day elevation, vehicle speed, battery power demand, and modeled cell temperature to communicate system behavior.

Miniature WALL-E Prototype (Ongoing) | *SolidWorks*

- Designed a fully articulated robotic hand in SolidWorks with 6 finger joints, using a rack-and-pinion driven sliding track for adjustable finger spacing to match WALL-E's grip.
- Derived spur gear and rack tooth geometry through analytical calculations from measured assembly clearances, optimizing pinion geometry to fit within a sub-8 mm envelope under tight packaging constraints.
- Optimized actuator placement by load location, using servo motors at proximal joints for lifting force and friction-fit joints distally, reducing actuator count and weight.
- Verified collision-free joint rotation and rack-pinion mesh clearance through SolidWorks interference detection and mate testing, catching geometry conflicts before committing to a print.